

CAKE: Confidence in Assignments via K-partition Ensembles

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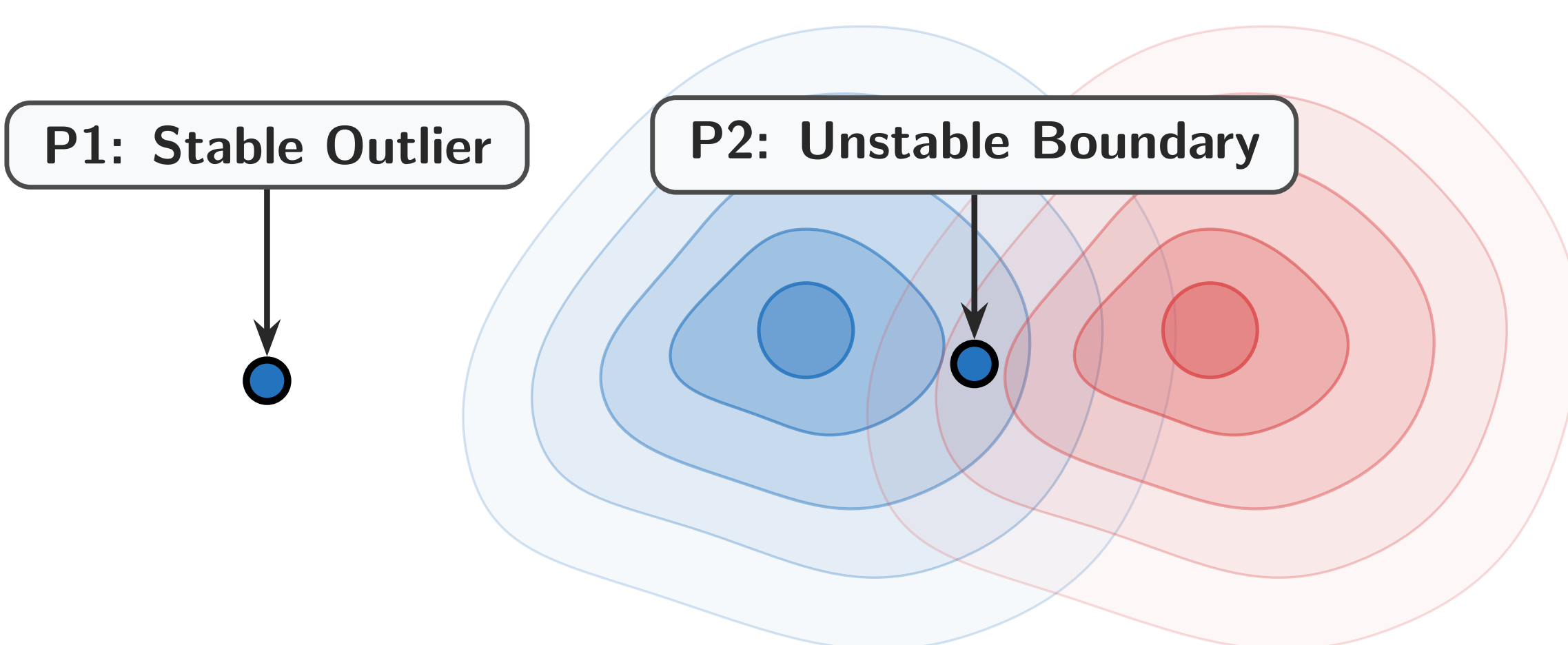
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1 Introduction

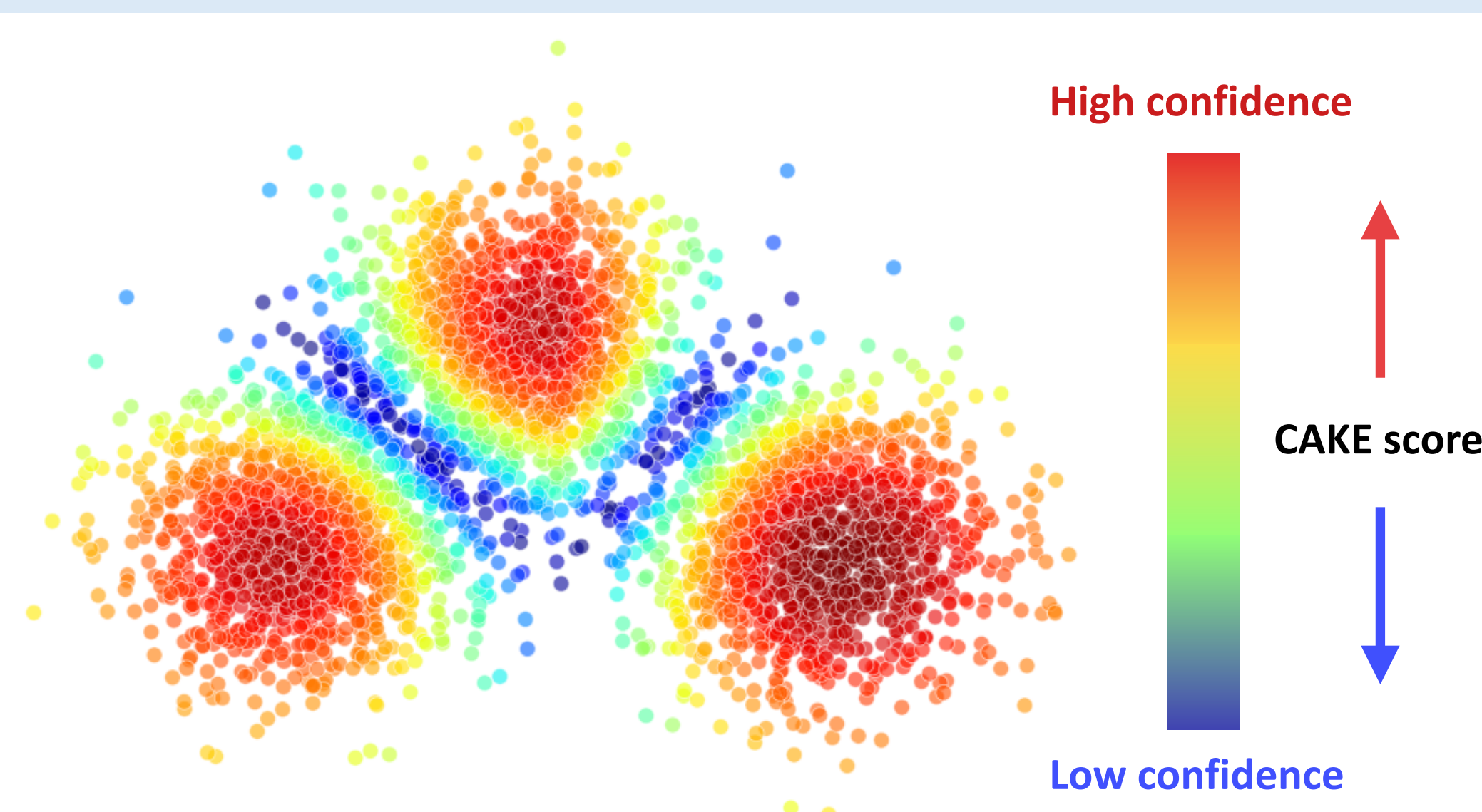
Clustering is widely used for unsupervised structure discovery, yet it provides limited insight into how reliable each individual assignment is. Diagnostics such as convergence behavior, objective values, or internal validation metrics may reflect global clustering quality, but they do not indicate whether particular instances are assigned confidently.

This limitation is especially important for initialization-sensitive algorithms such as k -means, where different random seeds can produce different partitions. Assignment-level instability can undermine both accuracy and robustness, especially when clustering results are used for downstream analysis, anomaly detection, or scientific discovery.

Existing ensemble approaches improve global consistency by aggregating multiple clustering runs, but they typically do not provide interpretable pointwise confidence scores. Agreement across runs alone can miss whether a point is geometrically well supported by its assigned cluster, while single-run geometric metrics can miss instability across different partitions.



We introduce **CAKE (Confidence in Assignments via K-partition Ensembles)**, a confidence estimation framework that assigns each data point an interpretable score reflecting both its assignment stability across clustering runs and the consistency of its geometric support under the learned clustering structure.



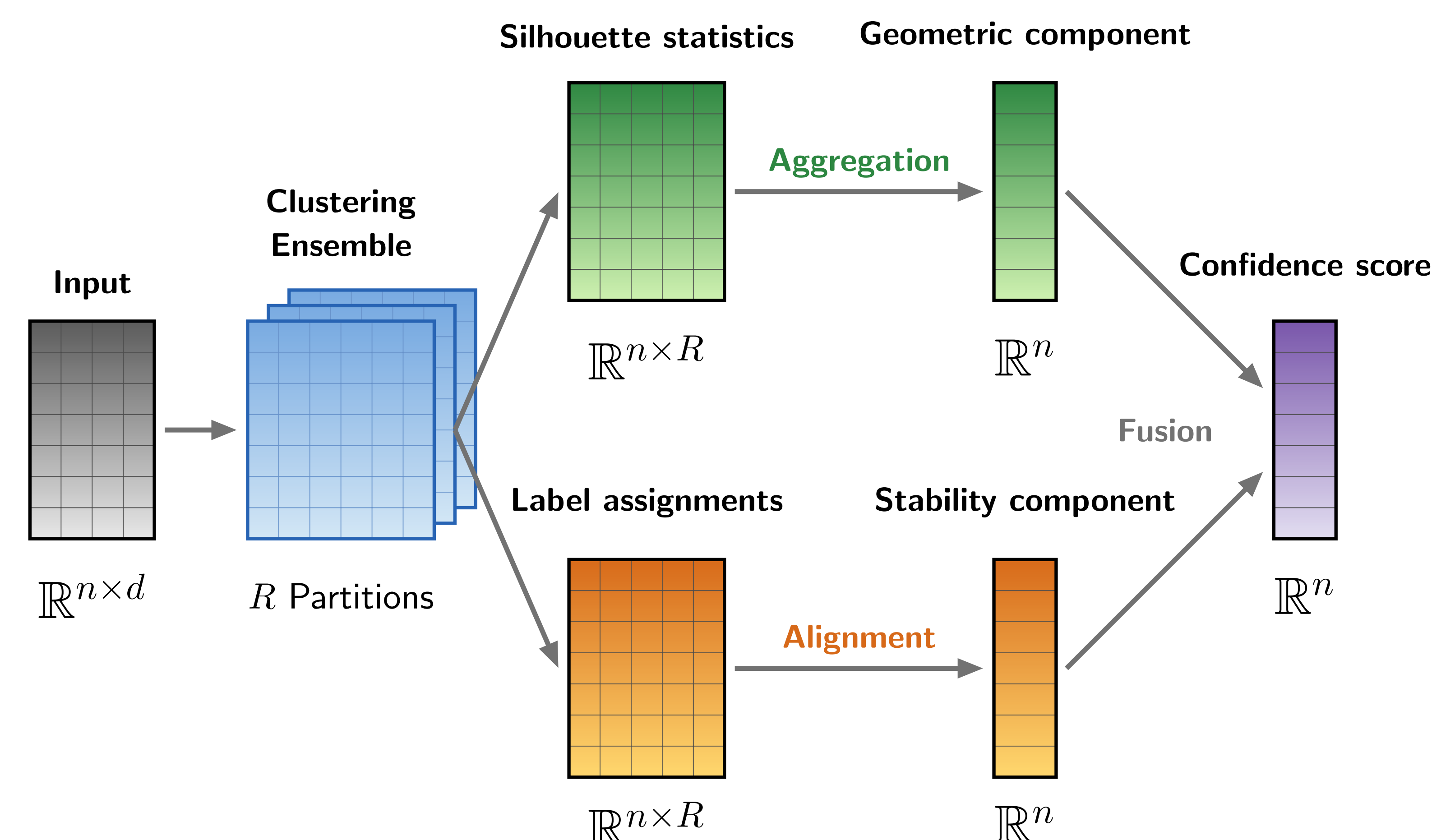
2 Methodology

CAKE estimates pointwise confidence using an ensemble of clustering partitions. The ensemble can be produced using one or more clustering algorithms, different random seeds, parameter settings, or resampling strategies, so that assignment variability can be observed.

For each data point, CAKE computes two complementary components:

Geometric consistency is computed as the mean Silhouette score of the point across the ensemble, penalized by its standard deviation across runs. It measures how well the point fits the cluster it is assigned to across the ensemble.

Assignment stability is computed as the fraction of clustering-run pairs in which the point receives the same assignment after label alignment. It measures how consistently the point receives the same cluster assignment across runs. Since cluster labels may be permuted between partitions, labels are first aligned using the Hungarian algorithm, and stability is then computed from pairwise agreement across aligned runs. The two components are fused into a single **confidence score** in $[0,1]$, either through a product or a harmonic mean formulation.



3 CAKE-curated clustering

CAKE-curated clustering yields higher-quality clusters than full-data clustering across synthetic and real-world datasets, with gains reaching up to 0.33 in ARI, 0.24 in AMI, and 0.20 in clustering accuracy. It outperforms alternatives based on random choice, stability-only signals, or geometry only signals.

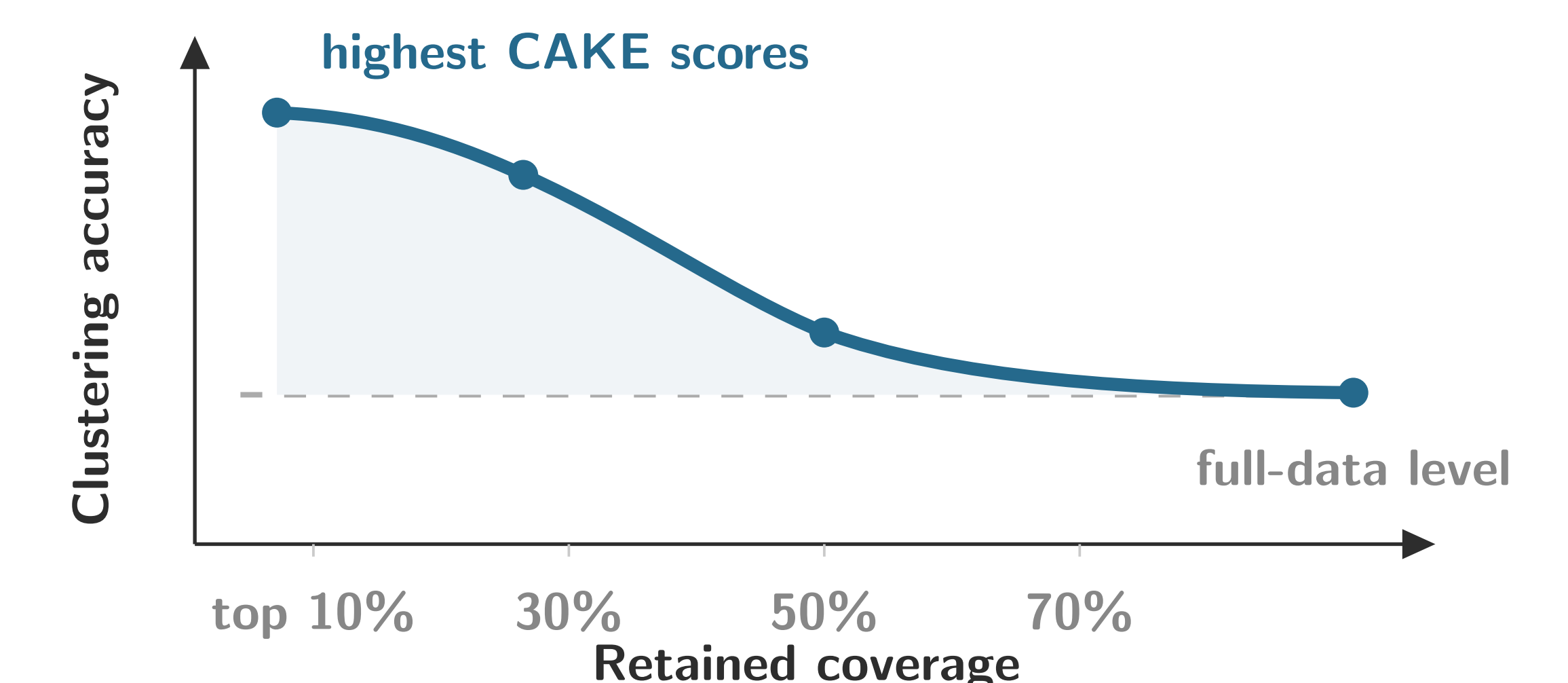
Mean clustering quality across agreement, information, and accuracy



4 Reliability ranking

Across synthetic and real-world datasets, CAKE produces an informative pointwise reliability ranking: the highest-scoring points concentrate correctly clustered instances, while lower-scoring points correspond to more ambiguous or borderline cases.

This ordering is also strongly reflected at the percentile level, with Spearman correlations between CAKE score percentiles and per-bin clustering accuracy ranging from 0.881 to 0.999 across synthetic and real-world datasets.



5 Assignment confidence

CAKE provides the most informative signal of consensus clustering quality against ensemble uncertainty, bootstrap stability, fuzzy memberships, and density-based alternatives. Across datasets, it achieves the highest mean AUPRC, which measures how well high-confidence scores identify correctly clustered instances, and the highest mean AUROC, which measures how well confidence separates reliable from unreliable assignments. Together, these results show that CAKE's combined stability-geometry signal is more reliable than confidence cues based only on uncertainty, membership, or density.

Confidence signal	Mean AUPRC	Mean AUROC
CAKE	0.885	0.800
FCM membership	0.832	0.735
FCM entropy	0.831	0.730
Bootstrap stability	0.811	0.619
Entropy agreement	0.790	0.576
Cluster-local LOF	0.785	0.598
Global LOF	0.770	0.567

Explore the paper!

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